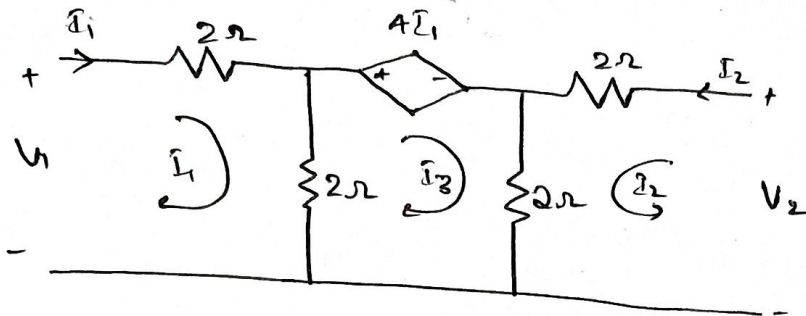


Two Port Network Parameters

I] Determine Z-parameters



Apply KVL to mesh 1

$$-2I_1 - 2(I_1 - I_3) + V_1 = 0$$

$$-4I_1 + 2I_3 + V_1 = 0$$

$$V_1 = 4I_1 - 2I_3 \quad \text{--- (1)}$$

Apply KVL to mesh 3

$$-2(I_3 - I_1) - 4I_1 - 2(I_3 + I_2) = 0$$

$$-2I_3 + 2I_1 - 4I_1 - 2I_3 - 2I_2 = 0$$

$$-2I_1 - 2I_2 - 4I_3 = 0$$

$$2I_1 + 2I_2 + 4I_3 = 0 \quad \text{--- (2)}$$

from eq (2)

$$4I_3 = -2I_1 - 2I_2$$

$$I_3 = \frac{-2I_1 - 2I_2}{4}$$

$$I_3 = -0.5I_1 - 0.5I_2 \quad \text{--- (a)}$$

Apply KVL to mesh 2

$$-2I_2 - 2(I_2 + I_3) + V_2 = 0$$

$$-4I_2 - 2I_3 + V_2 = 0$$

$$V_2 = 4I_2 + 2I_3 \quad \text{--- (3)}$$

Substitute I_3 value in eq (1) & (3)

$$\begin{aligned} \textcircled{1} \Rightarrow V_1 &= 4I_1 - 2I_3 \\ &= 4I_1 - 2(-0.5I_1 - 0.5I_2) \\ &= 4I_1 + I_1 + I_2 \\ V_1 &= 5I_1 + I_2 \quad \text{--- (4)} \end{aligned}$$

$$\begin{aligned}
 (2) \Rightarrow V_2 &= 4I_2 + 2I_2 \\
 &= 4I_2 + 2(-0.5I_1 - 0.5I_2) \\
 &= 4I_2 - I_1 - I_2 \\
 V_2 &= -I_1 + 3I_2 \quad \text{--- (5)}
 \end{aligned}$$

Z parameter equations.

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (6)}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (7)}$$

by comparing (4) & (5) with (6) & (7)
we get

$$Z_{11} = 5$$

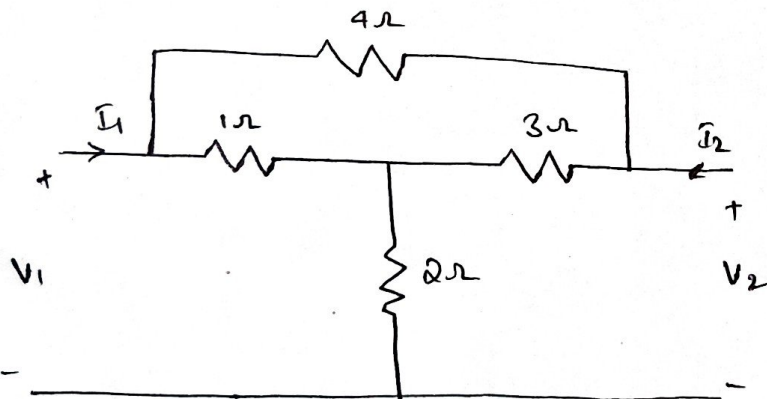
$$Z_{12} = 1$$

$$Z_{21} = -1$$

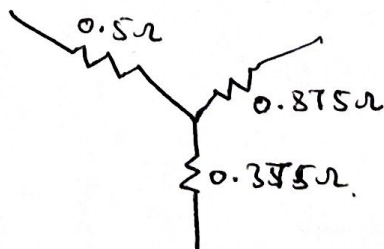
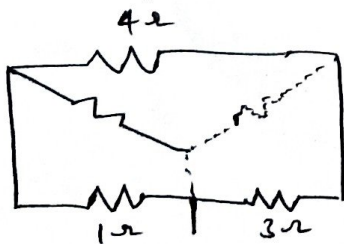
$$Z_{22} = 3$$

$$\underline{Z} = \begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix} \Omega$$

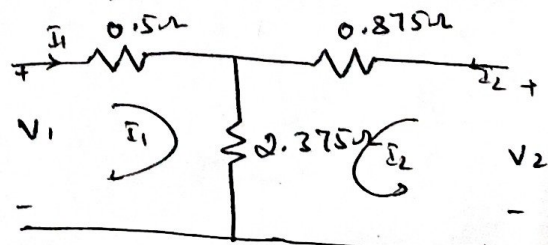
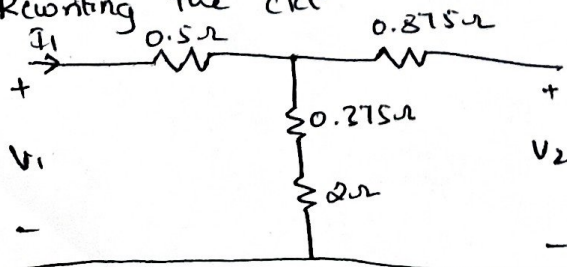
2] Find Z parameters for the network shown in figure.



converting delta to star



• Rewriting the ckt



Apply KVL to mesh 1

(3)

$$-0.5I_1 - 2.375(I_1 + I_2) + V_1 = 0$$

$$-0.5I_1 - 2.375I_1 - 2.375I_2 + V_1 = 0$$

$$V_1 = 2.875I_1 + 2.375I_2 \quad \text{--- (1)}$$

Apply KVL to mesh 2

$$-0.875I_2 - 2.375(I_2 + I_1) + V_2 = 0$$

$$-0.875I_2 - 2.375I_2 - 2.375I_1 + V_2 = 0$$

$$V_2 = 2.375I_1 + 3.25I_2 \quad \text{--- (2)}$$

Z parameters equations

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \quad \text{--- (3)}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \quad \text{--- (4)}$$

by comparing eq (1) & (2) with (3) & (4)
we get

$$Z_{11} = 2.875$$

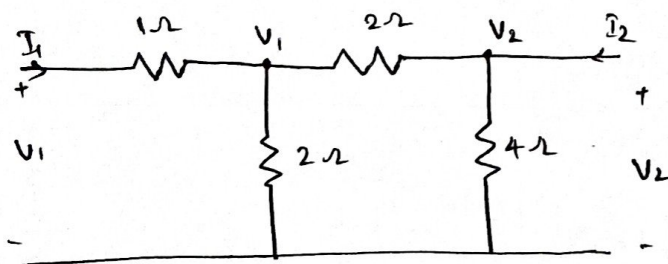
$$Z_{12} = 2.375$$

$$Z_{21} = 2.375$$

$$Z_{22} = 3.25$$

$$Z = \begin{bmatrix} 2.875 & 2.375 \\ 2.375 & 3.25 \end{bmatrix} \Omega$$

3] Determine Y parameters for the network



Apply KCL to node V_1

$$I_1 = \frac{V_1}{1} + \frac{V_1}{2} + \frac{V_1 - V_2}{2}$$

$$= V_1 + 0.5V_1 + 0.5V_1 - 0.5V_2$$

$$I_1 = 2V_1 - 0.5V_2 \quad \text{--- (1)}$$

Apply KCL to node V_2

$$I_2 = \frac{V_2}{4} + \frac{V_2 - V_1}{2}$$

$$I_2 = 0.25V_2 + 0.5V_2 - 0.5V_1$$

(4)

$$I_2 = -0.5V_1 + 0.75V_2 \quad \text{--- (2)}$$

Y parameter eq^{ns}

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \text{--- (3)}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \text{--- (4)}$$

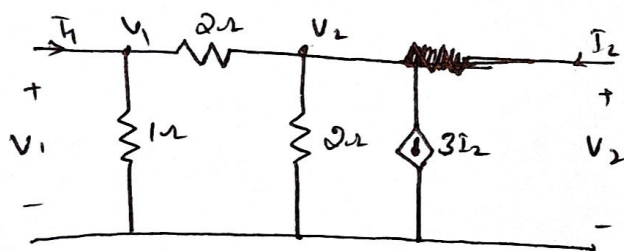
by comparing eqⁿ (1) & (2) with (3) & (4)

$$Y_{11} = 2 \quad Y_{12} = -0.5$$

$$Y_{21} = -0.5 \quad Y_{22} = 0.75$$

$$Y = \begin{bmatrix} 2 & -0.5 \\ -0.5 & 0.75 \end{bmatrix} \Omega$$

4] Find the network shown in Fig find Z & Y parameters



Apply KCL to node V_1

$$\frac{V_1}{1} + \frac{V_1 - V_2}{2} = I_1$$

$$I_1 = V_1 + 0.5V_1 - 0.5V_2$$

$$I_1 = 1.5V_1 - 0.5V_2 \quad \text{--- (1)}$$

Apply KCL to node V_2

$$I_2 - 3I_2 = \frac{V_2}{2} + \frac{V_2 - V_1}{2}$$

$$-2I_2 = 0.5V_2 + 0.5V_2 - 0.5V_1$$

$$-2I_2 = -0.5V_1 + V_2$$

$$I_2 = 0.25V_1 - 0.5V_2 \quad \text{--- (2)}$$

Y parameter eqⁿ

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \text{--- (3)}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \text{--- (4)}$$

By comparing eq (1) & (2) with (3) & (4) we get

(5)

$$y_{11} = 1.5 \text{ V}$$

$$y_{12} = -0.5 \text{ V}$$

$$y_{21} = 0.25 \text{ V}$$

$$y_{22} = -0.5 \text{ V}$$

$$Y = \begin{bmatrix} 1.5 & -0.5 \\ 0.25 & -0.5 \end{bmatrix} \Omega$$

Z parameters are given by transpose of Y

$$Z = Y'$$

$$Y' = \frac{\text{adj } Y}{\Delta Y}$$

$$Y = \begin{bmatrix} 1.5 & -0.5 \\ 0.25 & -0.5 \end{bmatrix}$$

$$\Delta Y = 1.5(-0.5) - (0.25)(-0.5)$$

$$= -0.75 + 0.125$$

$$= -0.625$$

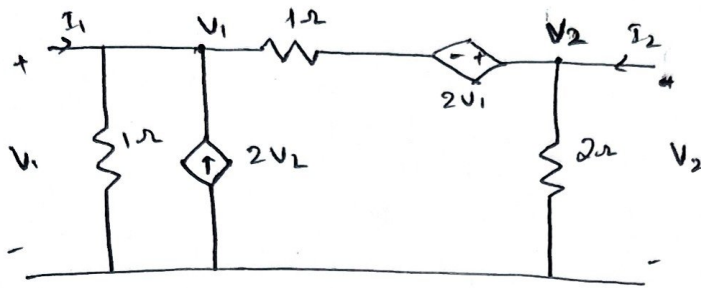
$$\text{adj } Y = \begin{bmatrix} -0.5 & 0.5 \\ -0.25 & 1.5 \end{bmatrix}$$

$$Z = \frac{\text{adj } Y}{\Delta Y} = \begin{bmatrix} \frac{-0.5}{-0.625} & \frac{0.5}{-0.625} \\ \frac{-0.25}{-0.625} & \frac{1.5}{-0.625} \end{bmatrix}$$

$$Z = \begin{bmatrix} 0.8 & -0.8 \\ 0.4 & -2.4 \end{bmatrix} \Omega$$

5] Find Y and Z parameters for the network.

6



Apply KCL at node V_1

$$I_1 + 2V_2 = \frac{V_1}{1} + \frac{V_1 + 2V_1 - V_2}{1}$$

$$I_1 + 2V_2 = 4V_1 - V_2$$

$$I_1 = 4V_1 - 3V_2 \quad \text{--- (1)}$$

Apply KCL at node V_2

$$I_2 = \frac{V_2}{2} + \frac{V_2 - 2V_1 - V_1}{1}$$

$$I_2 = 0.5V_2 + V_2 - 2V_1 - V_1$$

$$I_2 = -3V_1 + 1.5V_2 \quad \text{--- (2)}$$

Y parameter eq^{ns}

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \text{--- (3)}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \text{--- (4)}$$

By comparing (1) & (2) with (3) & (4) we get

$$Y_{11} = 4 \Omega$$

$$Y_{12} = -3 \Omega$$

$$Y_{21} = -3 \Omega$$

$$Y_{22} = 1.5 \Omega$$

$$Y = \begin{bmatrix} 4 & -3 \\ -3 & 1.5 \end{bmatrix} \Omega$$

$$Z = Y^{-1}$$

$$Y = \begin{bmatrix} 4 & -3 \\ -3 & 1.5 \end{bmatrix}$$

$$\Delta Y = 4(1.5) - (-3)(-3)$$

$$= 6 - 9$$

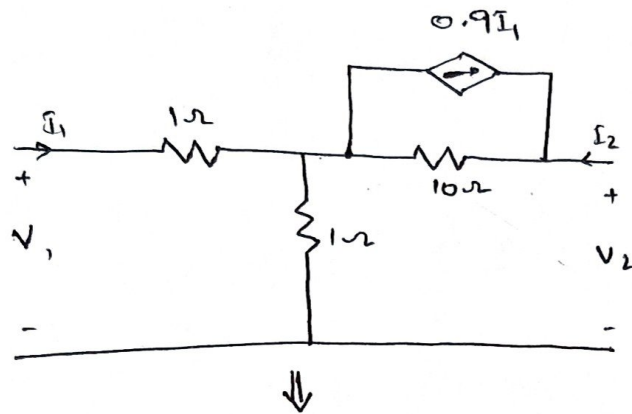
$$= -3$$

$$\text{adj } Y = \begin{bmatrix} 1.5 & 3 \\ 3 & 4 \end{bmatrix}$$

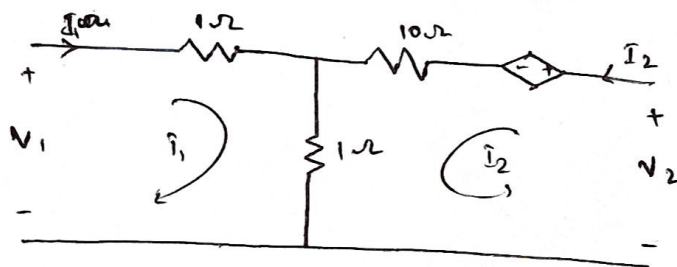
$$Z = \frac{\text{adj } Y}{\Delta Y} = \begin{bmatrix} 0.5 & -1 \\ -1 & -1.33 \end{bmatrix} \Omega$$

6] Find Z parameters for the network shown in figure 4 hence find h parameters

(7)



Step 1: Using source transformation the current source $0.9 I_1$ is converted to volt source as



Apply KVL to mesh 1

$$2 I_1 + I_2 = V_1 \quad \text{--- (1)}$$

Apply KVL to mesh 2

$$I_1 + 11 I_2 + 9 I_1 = V_2$$

$$10 I_1 + 11 I_2 = V_2 \quad \text{--- (2)}$$

Z parameters eqn

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (3)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (4)}$$

by comparing eqn (1) & (2) with (3) & (4) we get

$$Z_{11} = 2$$

$$Z_{12} = 1 \Omega$$

$$Z_{21} = 10$$

$$Z_{22} = 11$$

$$\underline{\underline{Z = \begin{bmatrix} 2 & 1 \\ 10 & 11 \end{bmatrix} \Omega}}$$

To write H -parameters

rewrite eq (2)

$$I_2 = \frac{1}{11} V_2 - \frac{10}{11} I_1 \quad \text{--- (5)}$$

⑧

$$V_1 = 2I_1 + I_2$$

$$V_1 = \frac{12}{11} V_1 + \frac{1}{11} V_2 \rightarrow (6)$$

$$V_1 = h_{11}I_1 + h_{12}V_2 \quad \text{--- (7)}$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \text{ --- (8)}$$

$$h_{21} = \frac{-10}{4}$$

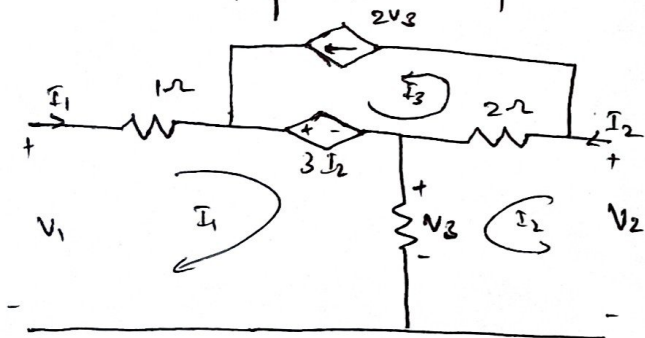
$$h_{22} = \frac{1}{11}$$

$$h_{11} = \frac{12}{11}$$

$$h_{12} = \frac{1}{11}$$

$$h = \begin{bmatrix} \frac{12}{11} & \frac{1}{11} \\ -\frac{10}{11} & \frac{1}{11} \end{bmatrix}$$

7) Find z & y parameters for the given ckt.



$$V_3 = 2(I_1 + I_2)$$

$$T_3 = 2V_3$$

$$= 2(2(I_1 + I_2))$$

$$= 4I_1 + 4I_2$$

$$-I_1 - 3I_2 - 2(I_1 + I_2) + V_1 = 0$$

$$V_1 = -I_1 - 3I_2 - 2I_3 - 2I_4$$

$$V_1 = 3I_1 + 5I_2 \text{ ——— (1)}$$

$$- 2(I_2 - I_3) - 2(I_1 + I_4) + V_2 = 0$$

$$-2\bar{I}_2 + 2\bar{I}_3 - 2\bar{C}\bar{I}_2 - 2\bar{U}_1 + \bar{V}_2 = 0$$

$$-2\mathbb{E}_2 + 2(4\mathbb{E}_1 + 4\mathbb{E}_2) - 2\mathbb{E}_2 - 2\mathbb{E}_1 + V_2 \geq 0$$

$$-2I_2 + 8I_1 + 8I_2 - 2I_2 - 2I_1 + V_2 = 0$$

⑨

$$V_2 = -6I_1 - 4I_2 \quad \text{--- (2)}$$

Z parameters eqn

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \quad \text{--- (3)}$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \quad \text{--- (4)}$$

by comparing eqn ① & ② with ③ & ④
we get

$$Z_{11} = 3 \quad Z_{12} = 5$$

$$Z_{21} = -6 \quad Z_{22} = -4$$

$$Z = \begin{bmatrix} 3 & 5 \\ -6 & -4 \end{bmatrix} \Omega$$

$$Y = Z^{-1}$$

$$Z^{-1} = \frac{\text{adj } Z}{\Delta Z}$$

$$Z = \begin{bmatrix} 3 & 5 \\ -6 & -4 \end{bmatrix}$$

$$\begin{aligned} \Delta Z &= 3(-4) - 5(-6) \\ &= -12 + 30 \\ &= 18 \end{aligned}$$

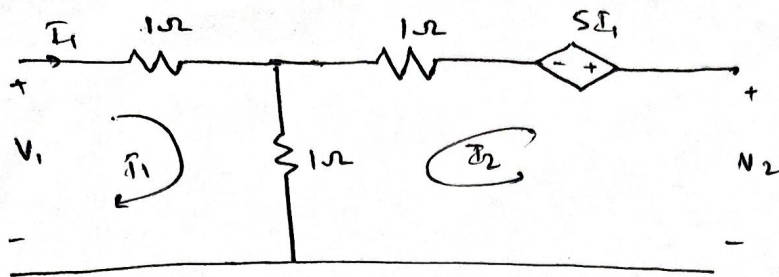
$$\text{adj } Z = \begin{bmatrix} -4 & -5 \\ 6 & 3 \end{bmatrix}$$

$$Y = \frac{\text{adj } Z}{\Delta Z} = \begin{bmatrix} -4/18 & -5/18 \\ 6/18 & 3/18 \end{bmatrix}$$

$$Y = \begin{bmatrix} -0.222 & -0.277 \\ 0.333 & 0.1667 \end{bmatrix} S$$

8] Determine Y-parameters

10



Apply KVL to mesh 1

$$-I_1 - 1(I_1 + I_2) + V_1 = 0$$

$$-I_1 - I_1 - I_2 + V_1 = 0$$

$$V_1 = 2I_1 + I_2 \text{ --- (1)}$$

Apply KVL to mesh 2

$$-5I_1 - I_2 - 1(I_2 + I_1) + V_2 = 0$$

$$-5I_1 - I_2 - I_2 - I_1 + V_2 = 0$$

$$V_2 = 6I_1 + 2I_2 \text{ --- (2)}$$

Z parameter equations

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \text{ --- (3)}$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \text{ --- (4)}$$

comparing eqⁿ (1) & (2) with (3) & (4)

$$Z_{11} = 2$$

$$Z_{12} = 1$$

$$Z_{21} = 6$$

$$Z_{22} = 2$$

$$Z = \begin{bmatrix} 2 & 1 \\ 6 & 2 \end{bmatrix} \Omega$$

For Y parameter

$$Y = \frac{\text{adj } Z}{\Delta Z}$$

$$\Delta Z = \begin{vmatrix} 2 & 1 \\ 6 & 2 \end{vmatrix}$$

$$= 2(2) - 6(1)$$

$$= -2$$

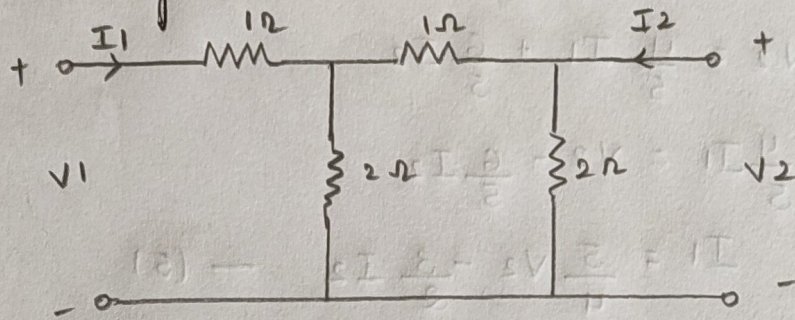
$$\text{adj } Z = \begin{bmatrix} 2 & -1 \\ -6 & 2 \end{bmatrix}$$

$$Y = \frac{\text{adj } Z}{\Delta Z} = \begin{bmatrix} 0.1 & 0.5 \\ 3 & 1 \end{bmatrix} S$$

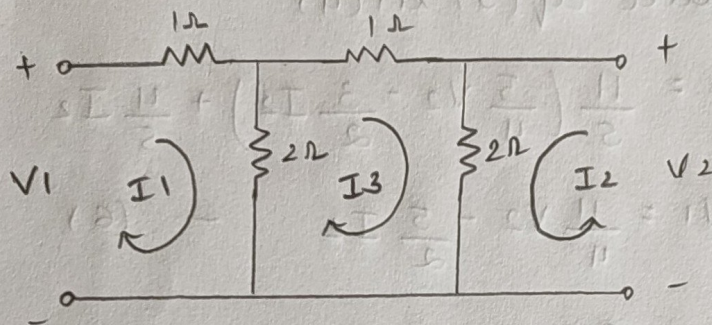
Module 3 :- Two ports Network Parameters.

(Unit -3)

1. Find the transmission parameter for the network shown in Fig :-



Solution :-



Applying KVL to mesh 1

$$V_1 = 3I_1 - 2I_3 \quad \text{--- (1)}$$

Applying KVL to mesh 2

$$V_2 = 2I_2 + 2I_3 \quad \text{--- (2)}$$

Applying KVL to mesh 3

$$\begin{aligned} -2(I_3 - I_1) - I_3 - 2(I_3 + I_2) &= 0 \\ -2I_3 + 2I_1 - I_3 - 2I_3 - 2I_2 &= 0 \end{aligned}$$

$$5I_3 = 2I_1 - 2I_2$$

$$I_3 = \frac{2}{5}I_1 - \frac{2}{5}I_2 \quad \text{--- (3)}$$

Substitute Eq (3) in (1)

$$V_1 = 3I_1 - 2\left(\frac{2}{5}I_1 - \frac{2}{5}I_2\right)$$

$$V_1 = 3I_1 - \frac{4}{5}I_1 + \frac{4}{5}I_2$$

$$V_1 = \frac{11}{5}I_1 + \frac{4}{5}I_2 \quad \text{--- (4)}$$

Substitute Eq (3) in (2)

$$V_2 = 2I_2 + 2 \left(\frac{2}{5} I_1 - \frac{2}{5} I_2 \right)$$

$$V_2 = 2I_2 + \frac{4}{5} I_1 - \frac{4}{5} I_2$$

$$V_2 = \frac{4}{5} I_1 + \frac{6}{5} I_2$$

$$\frac{4}{5} I_1 = V_2 - \frac{6}{5} I_2$$

$$I_1 = \frac{5}{4} V_2 - \frac{3}{2} I_2 \quad \text{--- (5)}$$

Substitute Eq (5) in (4)

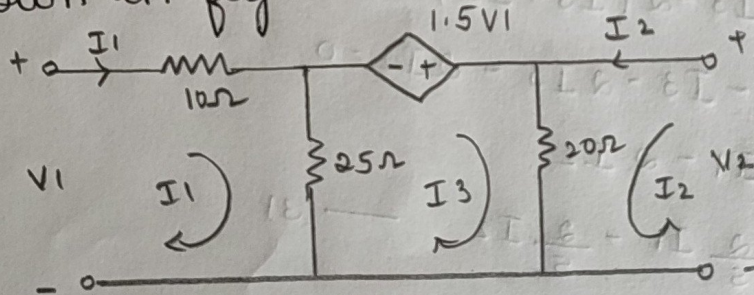
$$V_1 = \frac{11}{5} \left(\frac{5}{4} V_2 - \frac{3}{2} I_2 \right) + \frac{4}{5} I_2$$

$$V_1 = \frac{11}{4} V_2 - \frac{5}{2} I_2 \quad \text{--- (6)}$$

Comparing Eq (5) & (6),

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} \frac{11}{4} & -\frac{5}{2} \\ \frac{5}{4} & \frac{3}{2} \end{bmatrix}$$

2. Find the transmission parameter for the network shown in fig.



Apply KVL to mesh 1

$$V_1 = 10 I_1 + 25 (I_1 - I_3)$$

$$V_1 = 35 I_1 - 25 I_3 \quad \text{--- (1)}$$

Apply KVL to mesh 2

$$V_2 = 20(I_2 + I_3)$$

$$V_2 = 20I_2 + 20I_3 \quad \text{--- (2)}$$

Apply KVL to mesh 3

$$-25(I_3 - I_1) + 1.5V_1 - 20(I_2 + I_3) = 0$$

$$-25I_3 + 25I_1 + 1.5V_1 - 20I_2 - 20I_3 = 0 \quad (3)$$

$$-25I_3 + 25I_1 + 52.5I_1 - 37.5I_3 - 20I_2 - 20I_3 = 0$$

$$82.5I_1 = 77.5I_3 - 20I_2$$

$$I_3 = 0.94I_1 - 0.24I_2 \quad \text{--- (3)}$$

Substitute Eq (3) in Eq (1)

$$V_1 = 35I_1 - 25(0.94I_1 - 0.24I_2)$$

$$V_1 = 35I_1 - 23.5I_1 + 6I_2$$

$$V_1 = 11.5I_1 + 6I_2 \quad \text{--- (4)}$$

Substitute Eq (3) in (2)

$$V_2 = 20I_2 + 20(0.94I_1 - 0.24I_2)$$

$$V_2 = 20I_2 + 18.8I_1 - 4.8I_2$$

$$V_2 = 18.8I_1 + 15.2I_2 \quad \text{--- (5)}$$

$$18.8I_1 = V_2 - 15.2I_2$$

$$I_1 = 0.053V_2 - 0.81I_2 \quad \text{--- (6)}$$

Substitute Eq (6) in (4)

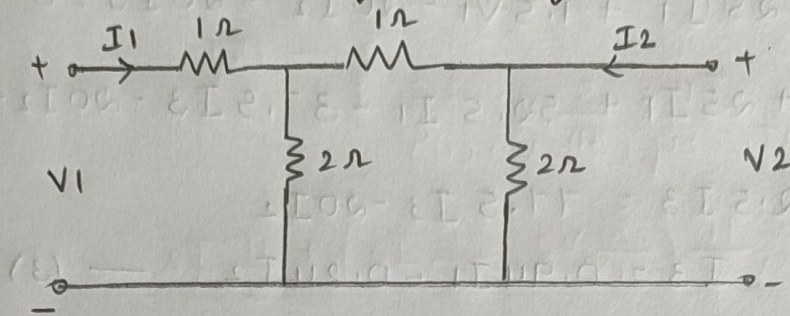
$$V_1 = 11.5(0.053V_2 - 0.81I_2) + 6I_2$$

$$V_1 = 0.61V_2 - 3.32I_2 \quad \text{--- (7)}$$

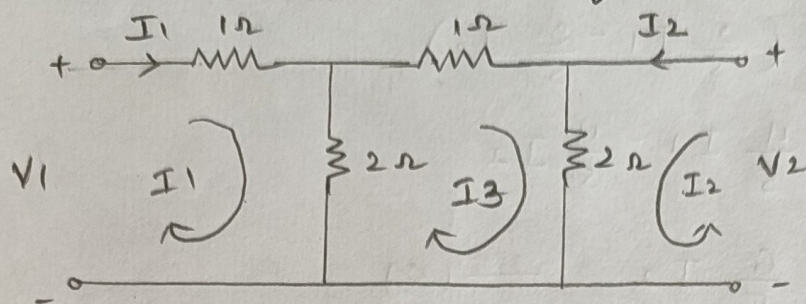
Comparing Eq (6) & (7), we get

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 0.61 & 3.32 \\ 0.053 & 0.81 \end{bmatrix}$$

3. Find h-parameter for the following circuits :-



3. Find h-parameter for the following circuit:



Apply KVL to mesh 1

$$V_1 = 3I_1 - 2I_3 \quad \text{--- (1)}$$

Apply KVL to mesh 2

$$V_2 = 2I_2 + 2I_3 \quad \text{--- (2)}$$

Apply KVL to mesh 3

$$-2(I_3 - I_1) - 1I_3 - 2(I_3 + I_2) = 0$$

$$-2I_3 + 2I_1 - 1I_3 - 2I_3 - 2I_2 = 0$$

$$2I_1 - 2I_2 - 5I_3 = 0$$

$$5I_3 = 2I_1 - 2I_2$$

$$I_3 = \frac{2}{5}I_1 - \frac{2}{5}I_2 \quad \text{--- (3)}$$

Substitute Eq (3) in (1)

$$V_1 = 3I_1 - 2\left(\frac{2}{5}I_1 - \frac{2}{5}I_2\right)$$

$$V_1 = 3I_1 - \frac{4}{5}I_1 + \frac{4}{5}I_2$$

$$V_1 = \frac{11}{5}I_1 + \frac{4}{5}I_2 \quad \text{--- (4)}$$

Substitute Eq (3) in (2)

$$V_2 = 2I_2 + 2\left(\frac{2}{5}I_1 - \frac{2}{5}I_2\right)$$

$$V_2 = 2I_2 + \frac{4}{5}I_1 - \frac{4}{5}I_2$$

$$V_2 = \frac{4}{5}I_1 + \frac{6}{5}I_2 \quad \text{--- (4)}$$

$$\frac{6}{5}I_2 = V_2 - \frac{4}{5}I_1$$

$$I_2 = \frac{5}{6}V_2 - \frac{2}{3}I_1$$

$$I_2 = -\frac{2}{3}I_1 + \frac{5}{6}V_2 \quad \text{--- (5)}$$

Substitute (5) Eq in (4)

$$V_1 = \frac{11}{5}I_1 + \frac{4}{5}\left(-\frac{2}{3}I_1 + \frac{5}{6}V_2\right)$$

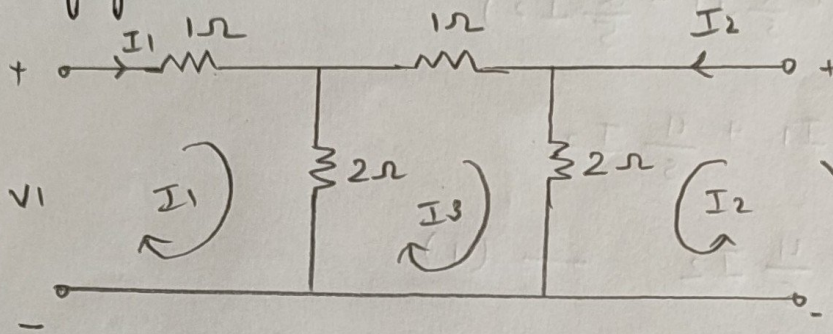
$$V_1 = \frac{11}{5}I_1 - \frac{8}{15}I_1 + \frac{20}{30}V_2$$

$$V_1 = \frac{25}{15}I_1 + \frac{20}{30}V_2 \quad \text{--- (6)}$$

Comparing Eq (5) & (6), we get

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{25}{15} & \frac{20}{30} \\ -\frac{2}{3} & \frac{5}{6} \end{bmatrix} //$$

4. Find the Z and Y parameter for the network shown in fig :-



Z - parameters are defined by :-

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Apply KVL to mesh 1

$$V_1 = 3I_1 - 2I_3 \quad \text{--- (1)}$$

Apply KVL to mesh 2

$$V_2 = 2(I_3 + I_2)$$

$$V_2 = 2I_3 + 2I_2 \quad \text{--- (2)}$$

Apply KVL to mesh 3

$$-2(I_3 - I_1) - 1I_3 - 2(I_3 + I_2) = 0$$

$$-2I_3 + 2I_1 - 1I_3 - 2I_3 - 2I_2 = 0$$

$$2I_1 - 2I_2 - 5I_3 = 0$$

$$5I_3 = 2I_1 - 2I_2$$

$$I_3 = \frac{2}{5}I_1 - \frac{2}{5}I_2 \quad \text{--- (3)}$$

Substitute Eq (3) in (1)

$$V_1 = 3I_1 - 2\left(\frac{2}{5}I_1 - \frac{2}{5}I_2\right)$$

$$= 3I_1 - \frac{4}{5}I_1 + \frac{4}{5}I_2$$

$$= \frac{11}{5}I_1 + \frac{4}{5}I_2 \quad \text{--- (4)}$$

Substitute Eq (3) in (2)

$$V_2 = 2I_2 + 2\left(\frac{2}{5}I_1 - \frac{2}{5}I_2\right)$$

$$\cancel{V_2 = 2I_2 + \frac{4}{5}I_1 - \frac{4}{5}I_2}$$

$$V_2 = 2\left(\frac{2}{5}I_1 - \frac{2}{5}I_2\right) + 2I_2$$

$$V_2 = \frac{4}{5}I_1 - \frac{4}{5}I_2 + 2I_2$$

$$V_2 = \frac{4}{5}I_1 + \frac{6}{5}I_2$$

Z-matrix (z):

$$[Z] = \begin{bmatrix} \frac{11}{5} & \frac{4}{5} \\ \frac{4}{5} & \frac{6}{5} \end{bmatrix} \Omega = \begin{bmatrix} 2.2 & 0.8 \\ 0.8 & 1.2 \end{bmatrix}$$

Y parameter:

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

Y parameter are found by inverting Z-matrix :-

$$[Y] = [Z^{-1}]$$

$$Z = \begin{bmatrix} 2.2 & 0.8 \\ 0.8 & 1.2 \end{bmatrix}$$

$$\Delta Z = (2.2 \times 1.2) - (0.8 \times 0.8)$$

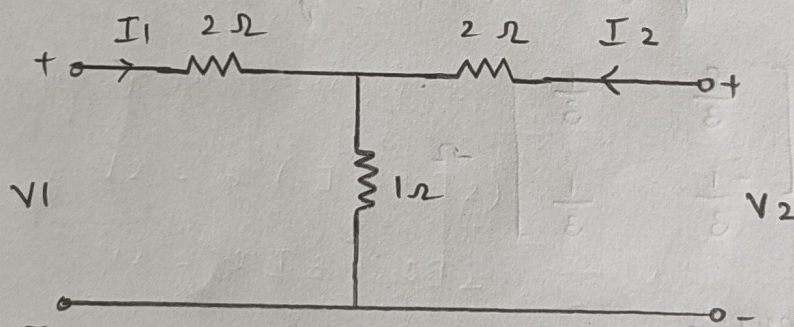
$$= 2.64 - 0.64$$

$$= 2.0$$

$$[Y] = \frac{1}{\Delta Z} \begin{bmatrix} 1.2 & -0.8 \\ -0.8 & 2.2 \end{bmatrix}$$

$$[Y] = \begin{bmatrix} 0.6 & -0.4 \\ -0.4 & 1.1 \end{bmatrix} \text{ //}$$

5. Determine h-Parameters:-



h-parameters:

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

Apply KVL to mesh 1

$$V_1 = 2I_1 + 1(I_1 + I_2)$$

$$V_1 = 2I_1 + I_1 + I_2$$

$$V_1 = 3I_1 + I_2 \quad \text{--- (1)}$$

Apply KVL to mesh 2

$$V_2 = 2I_2 + 1(I_1 + I_2)$$

$$V_2 = I_1 + 3I_2 \quad \text{--- (2)}$$

$$3I_2 = V_2 - I_1$$

$$h_{21} = -\frac{1}{3}$$

$$I_2 = -\frac{1}{3} \frac{V_2}{V_2} + \frac{1}{3} V_2 \quad \text{--- (3)}$$

$$h_{22} = \frac{1}{3}$$

Substitute Eq (3) in (1)

$$V_1 = 3I_1 + 1\left(-\frac{1}{3}I_1 + \frac{1}{3}V_2\right)$$

$$V_1 = 3I_1 - \frac{1}{3}I_1 + \frac{1}{3}V_2$$

$$h_{11} = \frac{8}{3}$$

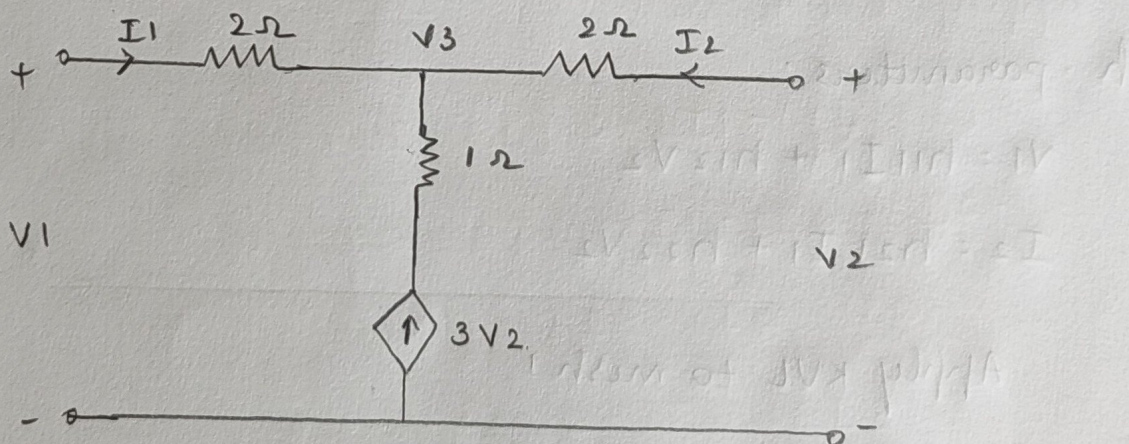
$$V_1 = \frac{8}{3}I_1 + \frac{1}{3}V_2 \quad \text{--- (4)}$$

$$h_{12} = \frac{1}{3}$$

On comparing (3) & (4), we get

$$[h] = \begin{bmatrix} \frac{8}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{bmatrix} \Omega$$

6. Find h-Parameters :-



$$I_1 = \frac{V_1 - V_3}{2}$$

$$= \frac{1}{2} V_1 - \frac{1}{2} V_3 \quad \text{--- (1)}$$

$$I_2 = \frac{V_2 - V_3}{2}$$

$$= \frac{1}{2} V_2 - \frac{1}{2} V_3 \quad \text{--- (2)}$$

Apply KCL at node 3

$$I_1 + I_2 + 3V_2 = 0 \quad \text{--- (3)}$$

Substituting Eq (1), (2) in (3)

$$\frac{V_1 - V_3}{2} + \frac{V_2 - V_3}{2} + 3V_2 = 0$$

$$2V_3 = V_1 + 7V_2$$

$$V_3 = \frac{1}{2} V_1 + \frac{7}{2} V_2 \quad \text{--- (4)}$$

Substitute Eq (4) in Eq (1),

$$I_1 = \frac{1}{2} V_1 - \frac{1}{2} \left(\frac{1}{2} V_1 + \frac{7}{2} V_2 \right)$$

$$= \frac{1}{4} V_1 - \frac{7}{4} V_2 \quad \text{--- (5)}$$

Substitute Eq (4) in (2),

$$I_2 = \frac{1}{2} V_2 - \frac{1}{2} \left(\frac{1}{2} V_1 + \frac{7}{2} V_2 \right)$$

$$= -\frac{1}{4} V_1 - \frac{5}{4} V_2 \quad \text{--- (6)}$$

comparing Eq (5), (6) we get

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & -\frac{7}{4} \\ -\frac{1}{4} & -\frac{5}{4} \end{bmatrix} //$$

7. Following short circuit currents and voltages are obtained experimentally, for a two port network:

Determine Y parameters

i. With output short circuited $I_1 = 5 \text{ mA}$, $I_2 = -0.3 \text{ mA}$,

and $V_1 = 25 \text{ V}$

ii. With input short circuited $I_1 = -5 \text{ mA}$, $I_2 = 10 \text{ mA}$,

$V_2 = 30 \text{ V}$.

$\Rightarrow$ For Y parameter:

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

i. with $V_2 = 0$

$$Y_{11} = \frac{I_1}{V_1} = \frac{5}{25} = 0.2 \times 10^{-3} \text{ S}$$

$$Y_{21} = \frac{I_2}{V_1} = \frac{-0.3}{25} = -0.012 \times 10^{-3} \text{ S}$$

ii. with $V_1 = 0$

$$Y_{12} = \frac{I_1}{V_2} = \frac{-5}{30} = -0.16 \times 10^{-3} \text{ S}$$

$$Y_{22} = \frac{I_2}{V_2} = \frac{10}{30} = 0.33 \times 10^{-3} \text{ S}$$

$$Y = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$$

$$Y = \begin{bmatrix} 0.2 \times 10^{-3} & -0.16 \times 10^{-3} \\ -0.012 \times 10^{-3} & 0.33 \times 10^{-3} \end{bmatrix} \text{ S}$$

8. Following are the hybrid parameters for a network
Determine the Y parameters for the network:-

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix}$$

$\Rightarrow$

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

Substitute, the values

$$V_1 = 5I_1 + 2V_2 \quad \text{--- (1)}$$

$$I_2 = 3I_1 + 6V_2 \quad \text{--- (2)}$$

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix}$$

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \quad \text{--- (3)}$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \quad \text{--- (4)}$$

$$I_1 = \frac{V_1 - 2V_2}{5}$$

$$I_1 = \frac{1}{5}V_1 - \frac{2}{5}V_2 \quad \text{--- (5)}$$

compare (5) & (3), we get

$$I_1 = 0.2V_1 - 0.4V_2$$

$$Y_{11} = 0.2$$

$$Y_{12} = -0.4$$

$$I_2 = 3 \left[\frac{V_1 - 2V_2}{5} \right] + 6V_2$$

$$I_2 = \frac{3}{5} V_1 - \frac{2}{5} V_2 + 6 V_2$$

$$= \frac{3}{5} V_1 + \frac{24}{5} V_2 \quad \text{--- (6)}$$

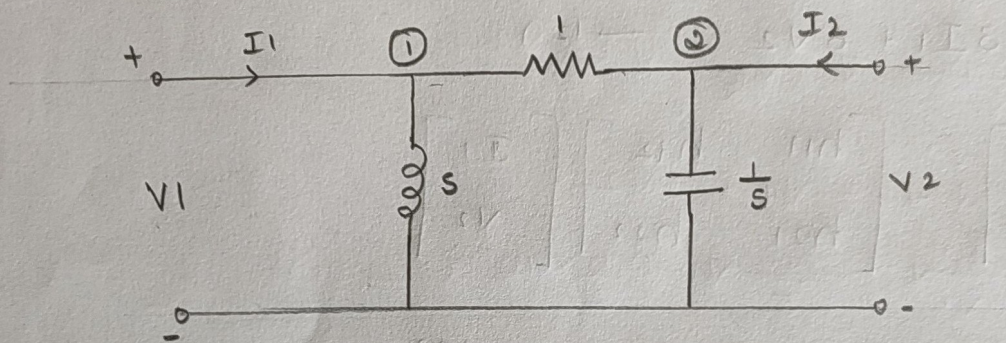
Compare (6) & (2), we get

$$Y_{21} = 0.6$$

$$Y_{22} = 4.8$$

$$= \begin{bmatrix} 0.2 & -0.4 \\ 0.6 & 4.8 \end{bmatrix}$$

9. Determine h-parameter :-



Apply KCL at node 1

$$I_1 = \frac{V_1}{5} + (V_1 - V_2)$$

$$= \frac{5+1}{5} V_1 - V_2 \quad \text{--- (1)}$$

Apply KCL at node 2

$$I_2 = \frac{V_2}{1} + (V_2 - V_1)$$

$$= (5+1) V_2 - V_1$$

$$= (5+1) V_2 - I_1 \quad \text{--- (2)}$$

Substitute Eq (ii) in Eq (i).

$$\begin{aligned}
 I_1 &= \frac{s+1}{s} [(s+1) V_2 - I_2] - V_2 \\
 &= \left[\frac{(s+1)^2}{s} - 1 \right] V_2 - \frac{s+1}{s} I_2 \\
 &= \left[\frac{s^2 + s + 1}{s} \right] V_2 - \left(\frac{s+1}{s} \right) I_2 \quad \text{--- (3)}
 \end{aligned}$$

Comparing Eq ii^o & ii^o , we get

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} s+1 & 1 \\ \frac{s^2 + s + 1}{s} & \frac{s+1}{s} \end{bmatrix} // \checkmark$$